Afzal Khan

My_Website | afzaljawadkhan@gmail.com | +92 337 9755627 | Afzal Khan

SUMMARY

AI Workflow & Data Engineer | RAG & Agentic Systems | MLOPS

Data and AI engineer with 4+ years building production RAG chatbots, multi-step agentic workflows, and clean data pipelines on Databricks and Snowflake. Blends traditional data engineering (SQL/Python, data auditing/cleaning, ETL automation) with practical LLM feature development and frontend work shipping clean chat widgets (popup bubbles and text cards). Daily user of AI-native coding tools (Cursor, Claude), with deep experience scripting data movement between platforms via secure OAuth connectors. Comfortable in fast-paced, fully remote teams on US Eastern hours.

EXPERIENCE

ClaimbAI — AI Engineer | Remote, California, USA (Jul 2025 – Present)

- Built a customer-support chatbot with complex **RAG** (indexing, chunking, embeddings) using Google LLMs via **BitNet**, so the AI pulls real-time, accurate answers with improved retrieval relevance and latency.
- Built and maintained 12+ secure **connectors** (Snowflake, Databricks, AWS, Azure, GitHub, Slack) with OAuth/RBAC, cleaning and organizing data tables to be AI-ready for reliable ingestion, vectorization, and RAG pipelines.
- Designed multimodal **voice agents** and agent workflows (voice→ASR→embeddings→agent), plus a local **SLM** for on-prem knowledge and private retrieval.
- Architected RAG infrastructure (PostgreSQL vector store + LlamaIndex) and automated **data pipelines**, enabling sub-second retrieval and reducing manual work by ~70%.
- Built clean **chat frontends** (popup bubbles and text cards) plus FastAPI REST APIs and CI/CD, using AI-native tools (Cursor, Claude) daily — improving deployment reliability and cutting rollback incidents ~40%.

COMCEPT — Software Engineer | Islamabad, Pakistan (Jan 2025 – Oct 2025)

- Engineered real-time computer vision and multi-object tracking stacks (YOLOv9–v11, GOTURN, CSRT, SAMURAI) and deployed optimized inference on NVIDIA Jetson Orin (aarch64).
- Optimized models via TensorRT, pruning, and quantization — improved inference throughput by **~30%** and reduced false positives by **25%**.
- Developed C++/Qt visualization and monitoring dashboards (40+ screens) to surface telemetry and reduce debugging time by **90%**
- Led system integration across firmware, Radxa/Armbian compute nodes, and networked devices; reduced system boot time and increased uptime by **95%.**
- Applied edge-deployment best practices (ONNX, aarch64 tuning) and performance profiling to meet latency and power constraints.

AIGENIUS — Chief Operating Officer (COO) | Islamabad, Pakistan (Jul 2024 – Dec 2024)

- Directed operations and recruitment while contributing technically; scaled engineering headcount and established hiring pipelines to support product roadmap.
- Led a major data transformation: converted a **200M-row** dataset across **~200** tables into a normalized 3NF data warehouse and developed star-schema semantic layers for analytics and forecasting.
- Built and deployed a Node.js/React agent chatbot and automated ETL using AWS Glue + SQL, improving dashboard refresh times and reducing manual ETL interventions by **4 hours**.
- Delivered proofs-of-concept including player tracking (OpenCV) and production scraping pipelines; improved client reporting cadence and decision-making through automated dashboards.

Convergent Business Technologies — Consultant, Analytics & Insights | Islamabad, Pakistan (Aug 2023 – Jul 2024)

- Implemented market-basket analysis (Apriori), dimensional modeling, and Tableau dashboards; automated refresh pipelines to reduce manual reporting workload.
- Performed ETL to transform raw sources into 3NF and built a dimensional model semantic layer for analytics consumption.
- Automated Pepsi KSA & UAE analytics project — reduced refresh cycle from 1 week → 3 days (57% reduction) and cut required team size from 6 → 2 through query tuning and scripting.
- Built ML pipelines for transcription and audio analytics (OCR + timestamped ASR + post-processing) for aviation prototyping, enabling time-aligned analytics and faster model iteration.

Upwork — Freelance Data Engineer and Developer | Remote (Aug 2022 – Aug 2023)

- Delivered end-to-end analytics and predictive models (churn, forecasting, inventory optimization) for telco, healthcare, and finance clients; owned data pipelines, feature engineering, model training, and deployment.
- Built automated ETL workflows using SQL and AWS Glue; decreased data latency and improved report freshness by 25%.
- Created Power BI / Tableau dashboards and scheduled reports for executive stakeholders; trained client teams on analytics best practices

to ensure adoption.

- Conducted statistical analyses using pandas, NumPy, and scikit-learn to derive actionable insights that informed client strategy.

PROJECTS

- **ADVANCED SPORTS PERFORMANCE ANALYTICS PLATFORM:** Architected and deployed predictive models for player positioning and performance metrics; delivered interactive dashboards used by coaching staff to improve tactical decisions and reduce match analysis time by 45%.
- **FACE RECOGNITION SOFTWARE:** Developed a high-performance facial recognition pipeline (MATLAB + transfer learning on AlexNet) achieving 92% classification accuracy on a custom dataset of 10,000+ images; optimized batch inference to improve throughput by 35%.
- **AI AGENT ECOSYSTEM FOR BUSINESS AUTOMATION:** Architected a multi-agent automation system across Sales, HR, Marketing, and Engineering; central controller orchestrated 10–15 specialized agents per department, automating repetitive tasks and reducing manual workload by ~60%.
- **ADAPTIVE LANGUAGE LEARNING WEB APP:** Firebase + Gemini API; LLM-curated lessons, conversational practice, and progress tracking.
- **LOCAL RESEARCH AI AGENT:** Local-only agent using n8n + Ollama + SERP API; used PostgreSQL vector DB for knowledge retrieval and persistent memory.
- **PERSONAL ASSISTANT AI AGENT**: Developed a personal assistant agent using OpenAI's API and n8n to manage calendars, Gmail, Telegram, WhatsApp, and other personal tasks.
- **CLAUDE DESKTOP MCP CLIENT:** Created a local MCP client for Claude Desktop, enabling the AI to add, remove, and read files from the local machine.

EDUCATION

MIT | MicroMasters, Statistics and Data Science Virtual

Courses: Probability - The Science of Uncertainty and Data, Statistics and Data Science

NUST | Mechatronics Islamabad, Pakistan

Bachelors, Mechatronics Engineering

CERTIFICATIONS

- McKinsey Forward Program | McKinsey
- Tableau Data Analysis | Philip Burton
- Google Data Analytics | Coursera
- Advanced SQL for data analysis | Udacity
- UI/UX Design | Calarts
- CS50 | Harvard

SKILLS & TECHNOLOGIES

- **AI / LLMs & Agents:** LangChain, LlamaIndex, RAG, Generative AI, Ollama, BitNet, Voice Agents, Agentic Workflows, n8n, Zapier, Make, GoHighLevel, Google ADK, Cursor, Claude
- **Machine Learning:** PyTorch, TensorFlow, Hugging Face, Transformer Models, YOLO (v9–v11), Whisper, Librosa
- **APIs & Integrations:** OpenAI, Anthropic, Gemini, REST/FastAPI, Zapier/Make, webhook-based integrations, OAuth/connectors (Snowflake, AWS, Azure, GitHub, Slack, email/calendar/tasks)
- **Model Optimization & Edge:** TensorRT, ONNX, quantization, pruning, NVIDIA Jetson Orin (aarch64), Edge AI, ARM64 tuning
- **MLOps & Cloud:** Docker, Kubernetes, FastAPI, MLflow, Firebase, Cloud Run, AWS Glue, Azure, Snowflake, Databricks, SageMaker
- **Data Engineering**: SQL, Python, PostgreSQL, Vector Databases, Snowflake, Databricks, Data Auditing & Cleaning, ETL, Data Pipeline Automation, 3NF & Dimensional Modeling
- **Analytics & Visualization:** Tableau, Power BI, Predictive Analytics
- **Programming:** Python, C++, Node.js, React, Qt
- **Additional Expertise:** Prompt Engineering, Model Deployment, API Integration, CI/CD